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MBAN 5570 – Accounting and Financial Analytics
Assignment II

Key Financial Ratios (2020 -2025) of Maple Leaf LLC.

Year	Gross Margin	Operating margin %	ROA %	Current ratio	Quick ratio	Debt-to-equity	Interest coverage (x)	DSO (days)	Inventory days	Asset Turnover
2020	30.95%	10%	7.11%	1.89	1.11	0.58	5.25	47.80	75.52	1.11
2021	29.67%	9%	6.03%	1.83	0.98	0.74	4.10	64.18	96.95	1.06
2022	28.00%	8%	4.84%	1.91	0.92	0.87	3.23	80.30	126.74	0.97
2023	25.93%	7%	2.76%	1.90	0.88	1.02	2.25	101.39	155.13	0.88
2024	26.50%	14%	6.32%	1.80	0.84	0.97	4.56	102.95	161.28	0.84
2025	21.67%	3%	-1.32%	1.70	0.73	1.09	0.69	112.54	182.50	0.79

Question 1: Financial Performance and Risk Diagnostics

A. Profitability

- Gross Margin declined from 30.95% from in 2020 to 21.67% in 2025. This indicates that Maple Leaf LLC is retaining a smaller portion of the revenue after covering cost of goods sold, consistent with the rising input/production cost, weaker pricing power, or reduced cost efficiency. It improved briefly in 2024 with 26.50% however this was not sustained into 2025, reinforcing overall deterioration in core gross profitability.

  (Gross margin, 2020) = Gross profit / Revenue = 130 / 420 = 30.95%

- Operating Margin decreased from 10% in 2020 to 3% in 2025, indication of a weak operation performance and reduced ability to generate operating income from sales. The sharp increase to 14% in 2024 followed by a steep reduction to 3% in 2025 suggests volatility and limited persistence of operating improvements, which warrants caution when assessing sustainable earnings capacity.

  (Operating Margin, 2020) = EBIT/Revenue or 42/420 =10%

Liquidity

- Current ratio declined from 1.89 in 2020 to 1.70 in 2025. The ratio remains above 1.0, which suggests that current assets still exceed current liabilities. However, the downward trend indicates that the short-term financial cushion is shrinking overtime.

  (Current ratio, 2020) = Total current assets/ Total current liabilities = 170/90 =1.89

- Quick Ratio declined more materially from 1.11 (2020) to 0.73 (2025). Because the quick ratio excludes inventory and focuses on near-cash resources (cash and receivables), the decline indicates weakening immediate liquidity and greater reliance on inventory liquidation or external financing to meet short-term obligations.

  (Quick ratio, 2020) = (Cash + Accounts Receivable/ Current Liabilities = (45+55)/90 =1.11

Solvency

- Debt -to-Equity increased from 0.58 in 2020 to 1.09 in 2025, indicating that Maple Leaf LLC has become more dependent with debt financing or growing shareholder funding overtime. A rising leverage profile increases financial risk particularly when profitability is weakening.

  (Debt-to-Equity) = (Short term debt+ long term debt)/ Total Equity = (20+90)/190=.58

- Interest Coverage declined from 5.25 (2020) to 0.69 (2025). The 2025 value below 1.0 implies operating income is insufficient to cover interest expense, representing a critical debt-servicing concern. Although coverage improved temporarily in 2024 (4.56x), the subsequent collapse in 2025 suggests an elevated refinancing or financial distress risk.

  (Interest Coverage) =EBIT/Interest Expense = 42/8 = 5.25

Operational Efficiency

- Day Sales Outstanding increased from 47.80 days (2020) to 112.54 (2025), indicating that receivables are being collected much more slowly. This deterioration suggests that weaker credit/collection effectiveness and a growing drag on cash conversion.

  DSO (days) = (Accounts Receivable/Revenue) x 365 = (55/420) x 365 = 47.80

- Inventory Days rose from 75.52 days from 2020 to 182.50 days in 2025, indicating that inventory is significantly slower. This trend increases likelihood of working – capital strain and may elevate the risk inventory obsolescence or discounting.

  Inventory days = (Inventory/[COGS]) x 365 = (60/ [-290]) x 365 = 75.52

B. Major trends over time that are improving, deteriorating, or volatile are as follows:

- The Profitability is deteriorating, with notable volatility. The significant decline in Gross Margin indicates the company is retaining less profit per dollar of sales, consistent with the Cost of Goods sold increasing faster than the revenue and/or weakening pricing power. As this happen, it reduces the gross profit “buffer” available to cover for the company’s SG &A, interest and taxes. Also, with the operating margin decline means

even after managing the operating expenses, the company's core operations generate less profit per dollar value. In 2024, the spike in the operating margin to 14% suggests a one-time cost improvement, however the sudden sharp drop to 3% indicates volatility rather than a stable improvement. Finally, it is also a major flag when ROA turned negative, from 7.11% in 2020 to -1.32% in 2025. This means that the company's assets such as Cash, Inventory, PPE are no longer generating positive returns. The Maple Leaf LLC business appears to tie up resources but not convert them into sustainable profit.

- Liquidity is deteriorating. The current ratio remains above 1.0 but declines from 1.89 (2020) to 1.70 (2025), indicating that while current assets still exceed current liabilities, the short-term financial cushion is narrowing. The quick ratio declines more sharply from 1.11 to 0.73, signaling reduced immediate liquidity because near-cash resources (cash and receivables) are increasingly insufficient to cover short-term obligations without relying on inventory conversion or additional financing. This pattern is consistent with growing working-capital strain and weaker short-term flexibility.

- Solvency is deteriorating. Debt-to-equity increases from 0.58 (2020) to 1.09 (2025), reflecting rising leverage and a greater dependence on debt financing relative to shareholder equity. This increases financial risk, particularly in a period of weakening profitability. Interest coverage declines from 5.25x to 0.69x, and the 2025 value below 1.0 indicates operating income is insufficient to cover interest expense. Although coverage improves temporarily in 2024 (4.56x), the sharp decline in 2025 suggests elevated debt-servicing pressure and heightened risk of refinancing constraints or financial distress.

- Operational efficiency deteriorates materially. Days sales outstanding (DSO) rises from 47.80 to 112.54 days, indicating substantially slower collections and increasing cash tied up in receivables. Inventory days increases from 75.52 to 182.50, suggesting significantly slower inventory turnover and heightened working-capital requirements, with added risk of markdowns or obsolescence. Asset turnover declines from 1.11 to 0.79, indicating reduced efficiency in generating revenue from the asset base. Together, these trends reflect a weakening cash conversion cycle and declining operational effectiveness.

- While the company showed temporary improvement such as the Debt-to-Equity's slight drop from 2023's 1.02 to 0.97 in 2024, signifying equity grew faster than debt that year which temporary reduced leverage, however with the overall 2020-2025 company performance remained to declined suggests that gains were not sustained.

C. Based on my above analysis, Maple Leaf LLC does not reflect or appear to be a financially healthy company by year 2025. The most decisive evidence is the combination of negative ROA (−1.32%) and interest coverage below 1 (0.69x), which indicates weakened profitability and inadequate

operating capacity to service debt. Another concern is the working-capital efficiency feeding solvency risk. DSO (112.54 days) and inventory days (182.50 days) show cash is increasingly trapped in operations, while interest coverage suggests the firm has limited room to absorb financing costs. This is a high-risk mix because operational cash strain often forces additional borrowing at the worst time.

Q2. Identification of Anomalies and Limitations of Ratio Analysis

A deeper review of the Maple Leaf LLC's financial statements reveals several anomalies that raise concerns about data integrity, earnings quality, and the reliability of ratio-based conclusions. These issues become evident only when the income statement, balance sheet, and the cash flow statement were examined together highlighting why ratio analysis must be supplemented with cross - statement validation and professional skepticism.

1. Profits not converting into Cash

Where It Appears	Why It Is Problematic (Financial/Accounting Risk)	Why Ratio Analysis Alone Misleads
IS vs. CFS: Net income positive in 2021–2023, but CFO = 2, –20, –35	Indicates revenue not collected in cash, operations are consuming cash suggests aggressive revenue recognition or poor working-capital management	Profitability ratios appear acceptable, masking the fact that the business is not self-funding

Although Maple Leaf reports positive net income in 2021–2023 (26, 25, and 17 million), cash from operations is almost nonexistent or negative (2, –20, –35 million). This pattern indicates that earnings are being generated through accruals rather than cash collection. The primary driver is the significant expansion in receivables and inventory, which absorbs cash and undermines the reliability of reported profits.

This issue is not visible through profitability ratios alone such as gross margin, operating margin, and ROA remain positive through 2024, creating the impression of a viable business. Only by reconciling the income statement with the cash flow statement does it become clear that the firm's operations are not self-funding and that earnings quality is deteriorating.

2. The Working capital is expanding faster than the Revenue

Where It Appears	Why It Is Problematic (Financial/Accounting Risk)	Why Ratio Analysis Alone Misleads
AR +236%, Inventory +292% vs. Revenue +43%; DSO and inventory days more than double	Signals deteriorating credit discipline, overproduction, or demand misalignment; traps cash and increases liquidity risk	Efficiency ratios show deterioration but cannot explain underlying operational or accounting drivers

Accounts receivable and inventory expand at multiples of revenue growth, reflected in sharply rising DSO and inventory days. This pattern points to operational strain showing slow collections, excess production, or weak demand forecasting. It also explains the negative CFO. Efficiency ratios highlight deterioration but cannot distinguish between operational issues, strategic decisions, or potential revenue-recognition concerns.

3. Rising leverage and Declining Debt- Servicing Capacity

Where It Appears	Why It Is Problematic (Financial/Accounting Risk)	Why Ratio Analysis Alone Misleads
Debt-to-equity rises to 1.09; interest coverage falls to 0.69x	Debt increasing while ability to service it is falling; indicates heightened solvency and refinancing risk	Debt-to-equity may appear manageable unless combined with cash-flow and interest-coverage context

Debt-to-equity nearly doubles while interest coverage falls below 1.0 by 2025. Combined with negative CFO, this indicates that the company is relying on debt to fund operations rather than growth. A ratio-only view of leverage could understate risk unless paired with cash-flow and interest-coverage analysis, which reveal a deteriorating solvency position.

4. Volatile One-Year spike in 2024 performance

Where It Appears	Why It Is Problematic (Financial/Accounting Risk)	Why Ratio Analysis Alone Misleads
Operating margin jumps to 14% then collapses to 3%; ROA follows same pattern	Suggests non-recurring items or accounting adjustments; improvement is not sustainable	A single-year ratio view may falsely imply operational recovery

The sudden improvement in operating margin and ROA in 2024, followed by an immediate collapse in 2025, suggests non-recurring or accounting-driven effects rather than structural improvement. A ratio-only analysis that over-weights 2024 could lead to overly optimistic conclusions about profitability and operational recovery.

5. Cross- Statement Inconsistencies

Where It Appears	Why It Is Problematic (Financial/Accounting Risk)	Why Ratio Analysis Alone Misleads
CFS "change in working capital" does not fully reconcile with BS movements	Indicates data integrity issues; undermines reliability of CFO and working-capital metrics	Ratios assume accurate data; inconsistent statements produce misleading conclusions

The "change in working capital" reported in the cash flow statement does not fully reconcile with year-over-year changes in AR, inventory, AP, and other current items. This raises data-integrity concerns and undermines confidence in CFO and liquidity assessments. Ratios cannot detect such inconsistencies; they assume the underlying data is accurate.

Why Financial Analysis Must go Beyond Ratios

While ratios provide useful snapshots of performance, they do not validate the underlying data or reveal deeper structural weaknesses in the financial statements. Maple Leaf's results demonstrate this clearly. Profitability can appear stable even as operating cash flow deteriorates, and rapid growth in working capital can hide emerging liquidity pressures. Leverage ratios may look acceptable despite a declining ability to service debt, and isolated improvements in a single year can distort the overall trend. In addition, inconsistencies across statements, such as mismatched working capital movements, show how data quality issues can undermine ratio-based conclusions. For these reasons, meaningful financial analysis requires more than calculating ratios. It depends on reconciling information across all statements, verifying cash flow patterns, and applying professional skepticism to ensure that conclusions reflect the company's true economic condition rather than surface-level metrics.

Q3. AI-Assisted Analysis, Validation, and Professional Judgement

A. AI Replication: Prompt and AI Output

Prompt Used

"Analyze Maple Leaf LLC's profitability, liquidity, solvency, and efficiency using the 2020–2025 financial statements. Identify anomalies or financial risks and comment on the company's overall financial health."

Summary of AI Output

The AI identified declining profitability, weakening liquidity, rising leverage, and deteriorating efficiency. It noted falling margins, declining current and quick ratios, increasing debt levels, and rising DSO and inventory days. It also flagged negative operating cash flow and working-capital strain. Overall, it concluded that Maple Leaf LLC shows signs of financial stress and may face liquidity and solvency challenges.

B. Where AI agreed with my analysis:

The AI's assessment aligned closely with several of my key findings. It correctly identified the decline in profitability, noting the downward trend in gross margin, operating margin, and return on assets. It also recognized weakening liquidity through falling current and quick ratios, and it accurately highlighted solvency concerns by pointing to rising leverage and deteriorating interest coverage. In terms of efficiency, the AI acknowledged the significant increase in both days sales outstanding and inventory days, consistent with my own analysis of slowing collections and rising inventory levels. Finally, it agreed that negative operating cash flow and mounting working-capital pressure represent major financial risks. Overall, the areas of agreement demonstrate that the AI captured the broad directional trends reflected in my ratio analysis and cross-statement review.

Where the AI Was Too Generic:

Although the AI captured the broad directional trends, much of its analysis remained high-level and lacked the specificity needed for meaningful financial interpretation. It noted that efficiency was deteriorating but did not quantify the severity of the changes, such as the doubling of days sales outstanding or the dramatic increase in inventory days. It also failed to compare the growth of receivables and inventory to the much slower growth in revenue, a critical indicator of operational strain. Several observations were presented in general terms without linking them to actual figures or explaining their implications. As a result, the AI's output read more like a surface-level summary than a detailed financial assessment, missing the depth required for professional analysis.

Where the AI Missed Financial Nuance:

Beyond being generic, the AI overlooked several important nuances that materially affect the interpretation of Maple Leaf's financial condition. It did not identify the intentional inconsistency in the working-capital reconciliation, a key data-integrity issue that undermines confidence in the cash-flow statement. It also failed to recognize the volatility of the 2024 performance spike or question whether the improvement was sustainable. The AI did not address the earnings-quality concern created by profits that do not convert into cash, nor did it highlight the disproportionate growth of receivables and inventory relative to revenue. Most critically, it did not connect negative operating cash flow with rising leverage, a combination that significantly elevates solvency risk. These omissions show that while AI can summarize trends, it does not reliably detect deeper structural issues or interpret the economic significance behind the numbers.

In several areas, the AI's conclusions were not only incomplete but also somewhat misleading. It described Maple Leaf's liquidity position as "adequate but weakening," which significantly understates the severity of the issue given that the quick ratio falls below one and operating cash flow is negative in multiple years. This characterization ignores the fact that the company is increasingly dependent on short-term debt to fund basic operations. The AI also failed to question

the reliability of the underlying data, even though the working-capital reconciliation contains inconsistencies that directly affect the accuracy of cash-flow reporting. In addition, it treated the financial trends as linear and did not consider the structural deterioration implied by the combination of rising leverage, falling interest coverage, and declining cash generation. By oversimplifying these issues, the AI's assessment risks giving a false sense of stability in areas where the company is experiencing significant financial strain.

In conclusion, I would not recommend Maple Leaf LLC to a client or investor at this time. Despite periods of acceptable reported profitability, the company's persistent negative operating cash flow, rapid and disproportionate growth in receivables and inventory, rising leverage, and weakening interest coverage indicate a fragile financial position. These issues, combined with data inconsistencies in the working-capital reconciliation, raise concerns about both earnings' quality and the reliability of the reported figures. Until Maple Leaf demonstrates sustained improvements in cash generation, tighter working-capital discipline, and a more conservative capital structure, the overall risk profile remains too high for a prudent investment recommendation.

Q4. Data integrity and consistency Issues

A. Data integrity and Consistency Issues

- Working-Capital reconciliation does not match the Balance Sheet movements
  The cash-flow statement reports a 2022 working-capital change of –65, but the balance-sheet movements imply a different figure. In 2022, accounts receivable increased by 30 (80 → 110), inventory increased by 40 (85 → 125), accounts payable increased by 5 (55 → 60), and other current liabilities increased by 2 (22 → 24). These movements translate to an expected working-capital outflow of –63, not –65. Even a small mismatch indicates that the CFO figure may not be fully reliable.

- Depreciation, capex, and PPE movements are not fully consistent
  Across 2020–2025, total depreciation expense is 126, and total capex is 195, implying a net PPE increase of roughly 69. However, PPE rises from 180 to 255, an increase of 75, with no disposals disclosed. This suggests incomplete PPE roll forward information or missing adjustments.

- Retained earnings do not reconcile with net income and dividends
  Retained earnings should follow beginning RE + net income - dividends, but several years do not align. In 2022, beginning retained earnings were 77, net income was 25, and dividends were 12, implying an ending balance of 90. Instead, retained earnings rise to 105, leaving an unexplained 15-million difference. This indicates potential omissions or misclassifications in equity movements.

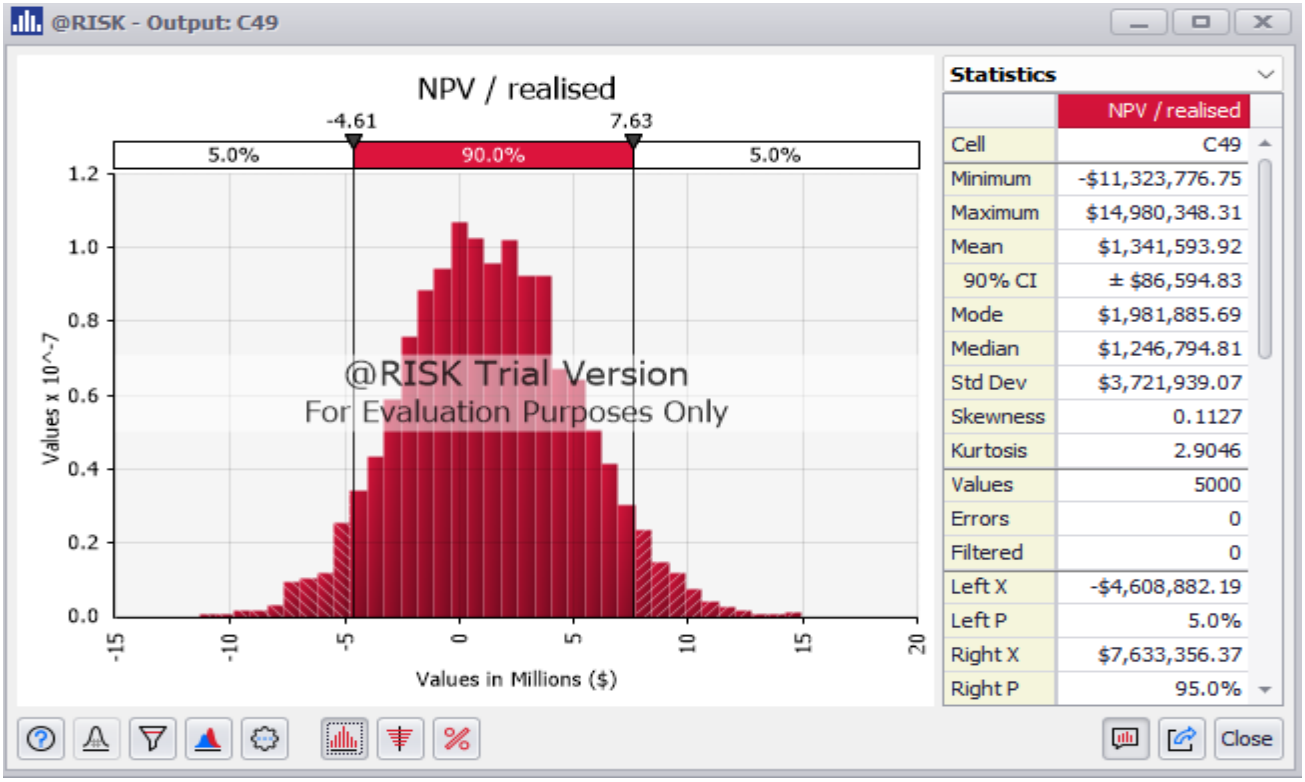
B. Why each issue represents a concern in accounting or financial analysis perspective
These inconsistencies undermine confidence in the financial statements. A mismatch in working-capital adjustments raises concerns about the accuracy of operating cash flow, which is central to liquidity and solvency analysis. Inconsistent PPE movements suggest that depreciation, Capital Expenditures, or asset disposals may not be properly recorded, affecting asset valuation and long-term investment analysis. Retained-earnings discrepancies imply that net income or dividends may be misstated, which directly affects profitability assessment and equity valuation. From an accounting perspective, these issues indicate incomplete accrual adjustments or missing supporting schedules. From an analytical perspective, they reduce the reliability of any ratio-based conclusions.

C. These data-quality problems can materially distort profitability, liquidity, and solvency assessments. Profitability ratios such as ROA, ROE, and net margin become unreliable if net income or equity balances are misstated. Liquidity and solvency analysis is compromised when working-capital movements do not reconcile, as CFO, current ratio, quick ratio, and debt-servicing capacity all depend on accurate cash-flow data. AI-assisted analysis is especially vulnerable because AI tools cannot detect reconciliation errors and will treat flawed inputs as accurate, leading to incorrect conclusions about financial health, risk, and performance trends.

D. Before using these statements for decision-making or AI-based analysis, Maple Leaf LLC should reconcile and restate all working-capital adjustments to ensure CFO reflects true operational performance. The company should also provide a complete PPE roll forward, including Capital Expenditures, depreciation, any disposals, and correct retained-earnings movements to ensure they align with net income and dividends. These steps are necessary to restore internal consistency and ensure that financial analysis is based on accurate, reliable data.

Part B.

Monte Carlo Simulation Summary



As part of the capital investment evaluation, I performed a Monte Carlo simulation using @RISK to quantify uncertainty in the project's Net Present Value (NPV). The simulation was run for 5,000 iterations on the NPV output cell, generating a probability distribution of outcomes under modeled demand, price, and cost variability.

Results indicate that the project has a mean simulated NPV of approximately $1.34M and a median of about $1.25M, with a standard deviation of roughly $3.72M, highlighting meaningful dispersion around the expected outcome. The simulated NPV range spans from approximately –$11.32M (minimum) to $14.98M (maximum).

From a downside-risk perspective, the 5th percentile (Value-at-Risk, 5% VaR cutoff) is approximately –$4.61M, meaning there is a 5% probability that the project's NPV will fall below –$4.61M. The central 90% of simulated outcomes lies between approximately –$4.61M and $7.63M, indicating that most results are positive but with a non-trivial probability of material loss.

Overall, the analysis suggests the project is expected to be value-creating on average, but decision-makers should explicitly consider the downside tail risk (low-probability, high-impact negative outcomes) when approving the investment.

Part C. Data Mining MACD Strategy

BuyDate	SellDate	BuyPrice	SellPrice	TradeProfit
2023-03-01 00:00:00	2023-03-04 00:00:00	100.2585	93.58939252	-6.66907762
2023-03-19 00:00:00	2023-04-03 00:00:00	92.92249	102.2601694	9.337683888
2023-04-11 00:00:00	2023-04-21 00:00:00	112.7541	110.0748119	-2.679305843
2023-05-21 00:00:00	2023-06-06 00:00:00	111.4145	114.3214678	2.907004185
2023-06-17 00:00:00	2023-06-23 00:00:00	124.4087	121.0462613	-3.362395741
2023-07-07 00:00:00	2023-07-10 00:00:00	123.7362	124.8569772	1.120798658
2023-07-18 00:00:00	2023-08-02 00:00:00	131.1335	133.1508898	2.017438668

MACD total profit (trade-by-trade)	Buy & Hold profit	Difference (MACD - BuyHold)	Number of completed trades
6.271306824	32.0107505	-25.73944367	7

1. MACD strategy total profit is $6.27 per share
2. Buy-and-hold profit: $32.01 per share
3. Better strategy and by how much: Buy-and-hold by $25.74 per share (MACD-B&H = -$25.74)

Using the MACD crossover rules and the required portfolio constraints (single-share, no short-selling, and mandatory initial purchase on 01-March-2023), the strategy generated a sequence of trades that produced a total profit of $6.27 per share over the period ending 11-September-2023. This value incorporates both realized gains from completed trades and the mark-to-market valuation of the final open position at the end of the sample.

In contrast, a passive buy-and-hold approach of purchasing one share on 01-March-2023 and holding it continuously until 11-September-2023, generated a substantially higher profit of $32.01 per share. The difference between the two approaches is therefore −$25.74 (MACD minus buy-and-hold), indicating that the MACD strategy underperformed buy-and-hold during this period.

The underperformance can be attributed to structural features of moving-average-based indicators. MACD signals are inherently lagging, causing entries to occur after upward trends have already begun and exits to occur after reversals are underway. In a period where prices generally moved upward, these delays reduce the strategy's ability to capture the full trend. Additionally, crossover reversals can produce "whipsaws," where the strategy exits and re-enters the position without meaningful price movement, reducing cumulative returns. Because the strategy is constrained to hold at most one share and cannot short the asset, it cannot offset these effects through contrarian positioning. Overall, the results suggest that over this sample window, buy-and-hold outperformed the MACD rule-based strategy.